Limiting Reactants and Percent Yield

Limiting and excess reactants, theoretical yield and percent yield

Name _____ Class _____ Date _____

DATA

H = 1.008 g/mol C = 12.01 g/mol O = 16.00 g/mol Al = 26.98 g/mol Cl = 35.45 g/mol

Convert every reactant to moles first. Theoretical yield comes from the limiting reactant.

A. Understanding

1. Define (a) limiting reactant, (b) excess reactant and (c) theoretical yield.

2. Give two reasons why the percent yield of a reaction is usually less than 100%.

B. Working in moles

3. For $2\,H_2 + O_2 \longrightarrow 2\,H_2O$, 3.0 mol of H_2 is mixed with 2.0 mol of O_2. (a) Which is limiting? (b) How many moles of water form? (c) How many moles of the excess reactant remain?

4. For $N_2 + 3\,H_2 \longrightarrow 2\,NH_3$, 1.5 mol of N_2 is mixed with 3.0 mol of H_2. (a) Which is limiting? (b) How many moles of ammonia form? (c) How many moles of the excess reactant remain?

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C. Working in grams

5. Aluminium reacts with chlorine: $2\,\mathrm{Al} + 3\,\mathrm{Cl_2} \longrightarrow 2\,\mathrm{AlCl_3}$. 5.40 g of aluminium is mixed with 15.0 g of chlorine. (a) Which reactant is limiting? (b) What mass of $\mathrm{AlCl_3}$ can form? (c) What mass of the excess reactant is left over?

6. In the experiment in question 5, 15.2 g of $\mathrm{AlCl_3}$ is actually collected. Calculate the percent yield.

7. Methane burns: $\mathrm{CH_4} + 2\,\mathrm{O_2} \longrightarrow \mathrm{CO_2} + 2\,\mathrm{H_2O}$. 10.0 g of methane is mixed with 32.0 g of oxygen. (a) Which is limiting? (b) What mass of $\mathrm{CO_2}$ forms? (c) What mass of the excess reactant is left over?

D. Challenge

8. A student reports a percent yield of 108%. Suggest a likely reason, and explain why a yield above 100% is impossible for a pure, dry product.

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